Ameena Khan

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PROFILE

AI Developer with expertise in Natural Language Processing (NLP), Large Language Models (LLMs), and Machine Learning. Skilled in Python, Ruby on Rails, PostgreSQL, MySQL, and Vector Databases, with experience in developing LLM-based applications, text-to-speech systems, and OCR-driven automation. Proficient in ReactJS and VueJS for frontend development. Passionate about building intelligent, scalable solutions and optimizing data processing workflows. Strong problem-solving, teamwork, and leadership skills, committed to innovation and continuous learning.

WORK EXPERIENCE

Artificial Intelligence Technology Center (AITeC), National Center for Physics (NCP) May 2024 – present | Islamabad, Pakistan
Assistant Manager Technical (AM Tech)/ AI Developer

Cyber Intelligent BOT

Technologies: Python, LangChain, Large Language Models, Retrieval-Augmented Generation, Vector Database, PostgreSQL, Streamlit, PyTorch, OCR, ROS 2, Jetson Xavier NX, Ubuntu 22.04

- Developed a secure Retrieval-Augmented Generation chatbot with general conversational capabilities powered by a locally hosted, pre-trained offline Large Language Model.
- Implemented document-based question answering for text-based and scanned PDFs, images, and other documents using integrated OCR, text extraction, chunking, embedding, and semantic retrieval pipelines.
- Conducted a research-driven evaluation of eight open-source and commercial OCR models for information extraction from noisy and degraded documents, identifying the best-performing model across public and custom datasets.
- Designed a local vector-database architecture to preserve data privacy and sovereignty, enabling offline document retrieval and conversational AI without dependence on external cloud services.
- Integrated PostgreSQL for user management, application data, and persistent chat functionality.
- Deployed the offline chatbot within a ROS 2 Foxy-based environment on an NVIDIA Jetson Xavier NX platform running Ubuntu 22.04.
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Urdu LLM Fine-Tuning

Technologies: Python, PyTorch, Hugging Face Transformers, Parameter-Efficient Fine-Tuning, LoRA, Urdu NLP, Streamlit, NVIDIA GPU

- Fine-tuned and evaluated open-source Large Language Models on Urdu instruction and conversational datasets to improve Urdu-language understanding, response generation, and domain-specific performance.
- Applied parameter-efficient fine-tuning techniques, including LoRA, to reduce computational and memory requirements while adapting the models for Urdu NLP tasks.
- Developed an interactive inference application for testing Urdu prompts and comparing the fine-tuned model with the original base model using qualitative and performance-based evaluation criteria.

EDUCATION

MS Computer Science, *PIEAS University, Nilore* Dec 2021 – Oct 2023 | Islamabad
CGPA 3.59/4

MSc Computer Science, *University of the Punjab (PUCIT)* 2019 – 2021 | Lahore, Pakistan
- CGPA 3.80/4, Gold Medalist

SKILLS

Backend Languages
Ruby on rails, Python

Soft Skills
Problem Solving, Team Work, Leadership

Databases
Postgres, MySQL, Vector Database

Artificial Intelligence
Natural Language Processing, Large Language Models (LLM), Machine Learning Algorithms, Deep Learning, Computer Vision Libraries: Pytorch, numpy, pandas, Scipy

JavaScript
ReactJS, VueJS, TypeScript

CERTIFICATES

CLE-Post Graduate Certificate
Certificate in Text To Speech Synthesis & Speech Corpus Development

Certificate of Organizer
Organized and conducted AI Innovative Ideas Competition **(1st AIIC-2024)** and worked on the registration committee for the National Artificial Intelligence Seminar.
- **NAIS-2024**
- **NAIS-2025**

Certificate of Instructor in Advanced ML&DL and Generative AI Course
Instructor of Generative AI and Prompt Engineering
AITeC/NCP | 2025

AWARDS

Gold Medalist, *University of the Punjab* Jun 2021
Awarded for securing **1st position** in the MSc (2019–2021) with a **CGPA of 3.80/4.00**

PUBLICATIONS

Deepfake News Detection: A Multimodal Framework Integrating LipNet, DeepSpeech and ResNET for Enhanced Audio-Visual Analysis 22 Jul 2026

Ameena Khan, Muhammad Ahsan Aziz, Muhammad Junaid Asif, Naeem Akhter, Rana Fayyaz Ahmad

- This paper proposes a multimodal deepfake news detection framework that combines lip movements, speech, and facial features extracted using LipNet, DeepSpeech2, BlazeFace, and ResNet18.
- Evaluated on the FakeAVCeleb dataset, the framework achieved 94% accuracy and outperformed a state-of-the-art multimodal ensemble baseline.